

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/M_DEB_SAAWKZe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 33344, C2 = 3663, C3 = 12552)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,      0.2, 0.01,      0.8);
Distrib (pAm,         TruncNormal_cv,       100,      0.2, 40,      2000);
Distrib (r_pAm_pM,    TruncNormal_cv,       0.8,      0.1, 0.1,      2);
Distrib (v,           TruncNormal_cv,       0.18505,  0.1, 0.001,  2);
Distrib (kap,         Uniform,               0,        1);
Distrib (kapA,        Uniform,               0.1,      0.9);
Distrib (Ehp,         TruncNormal_cv,       100,      0.2, 40,      300);
Distrib (E_coc,       TruncNormal_cv,       30,      0.2, 1,      300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001,  0.3);
Distrib (K,           Uniform,               0.00001,  2);
Distrib (z,           Uniform,               0,        1);
Distrib (dv,          TruncNormal_cv,       2,      0.5, 0,      10);
Distrib (wE,          LogUniform,           0.000000001, 0.1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,      Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter, RTOL, ATOL)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	Fm.1.	pAm.1.	r_pAm_pM.1.	v.1.	kap.1.	kapA.1.
Result_mode	0.48891900	142.23600	0.8981830	0.17840200	0.97014000	0.72909900
2.5%	0.35636910	93.32980	0.7556870	0.15595100	0.93828530	0.58234525
97.5%	0.63829090	160.48700	1.0298700	0.22278610	0.97780800	0.78737115
Min	0.10625500	69.97460	0.6483210	0.13630100	0.13646300	0.52275100
Max	0.75644300	170.51400	1.1657100	0.24942200	0.98290900	0.84342700
Result_mean	0.49010945	128.62345	0.8974464	0.18831007	0.96062561	0.68825705
Result_sd	0.07389728	17.18079	0.0695603	0.01783154	0.01084993	0.05413441

	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	K.1.	z.1.
Result_mode	102.73300	35.104900	0.0000172857	0.49432100	0.5517470000
2.5%	76.93430	21.944870	0.0000117163	0.33586800	0.1168932500
97.5%	147.75125	43.694700	0.0059954815	0.59700800	0.6350786000
Min	60.91520	15.071100	0.0000100145	0.26679000	0.0000690166
Max	178.27900	52.248200	0.0088419400	0.98193800	0.7536020000
Result_mean	111.92522	33.306588	0.0013484558	0.48357108	0.4715470235
Result_sd	18.01047	5.531718	0.0018719783	0.06337692	0.1227416764

	dv.1.	wE.1.	Sigma_W.1.
Result_mode	1.4153400	1.154550e-05	0.10042000
2.5%	1.2558610	8.754340e-06	0.09480410
97.5%	2.1797425	2.035790e-05	0.12024300
Min	1.0807400	9.636700e-09	0.08519830
Max	2.8634400	2.219700e-05	6.00633000
Result_mean	1.6237672	1.325938e-05	0.10680878
Result_sd	0.2358181	2.840104e-06	0.05624447

Number of observations, n = 277

Number of parameters, k = 13

Log-likelihood, LL = 11.22786

BIC = 50.65651

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.10	1.30
E_coc.1.	1.09	1.25
Ehp.1.	1.05	1.17
Fm.1.	1.27	1.76
K.1.	1.51	2.32
kap.1.	1.13	1.37

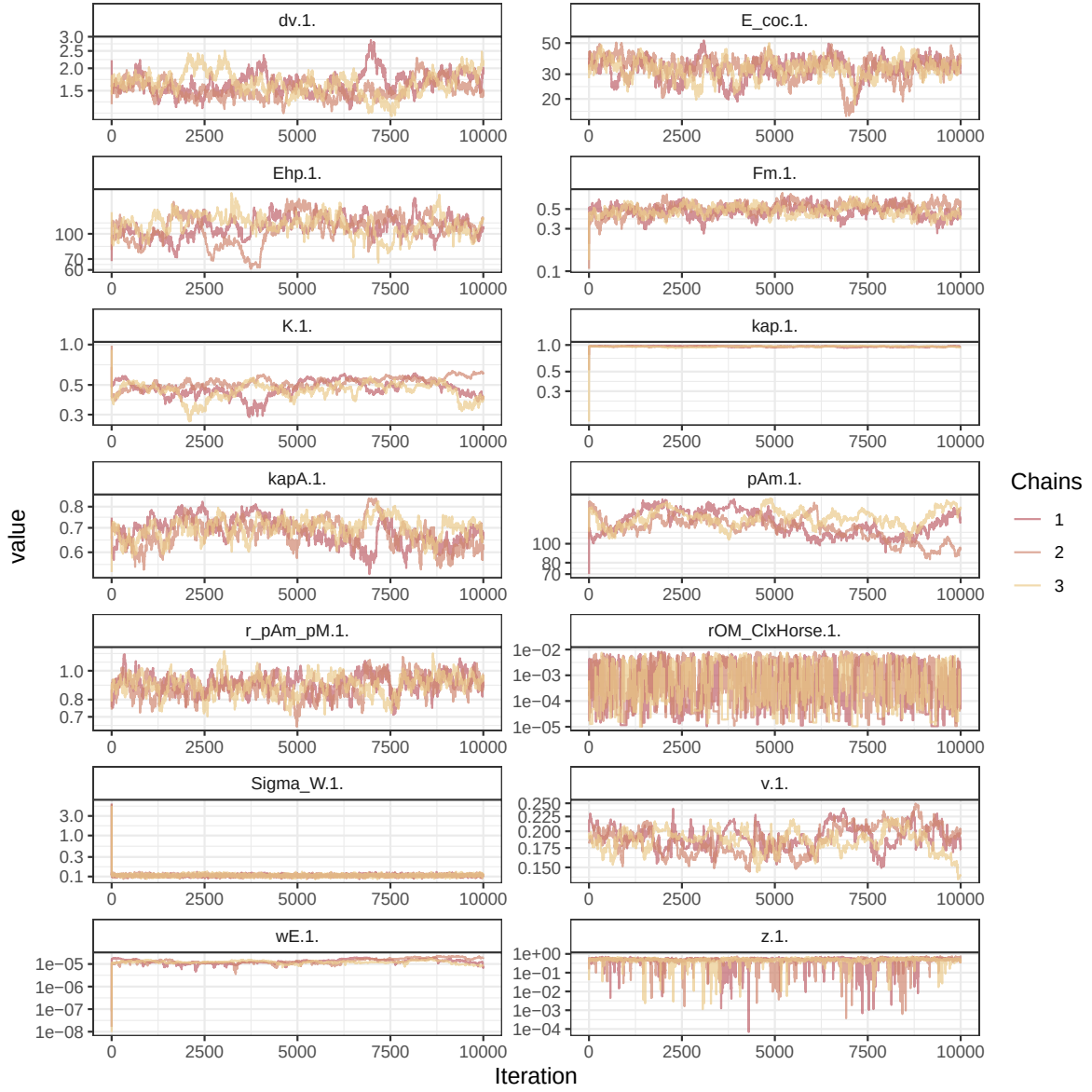


Figure 1: Parameters chains. Only the first iteration and the last `Nb_iter_kept` are represented.

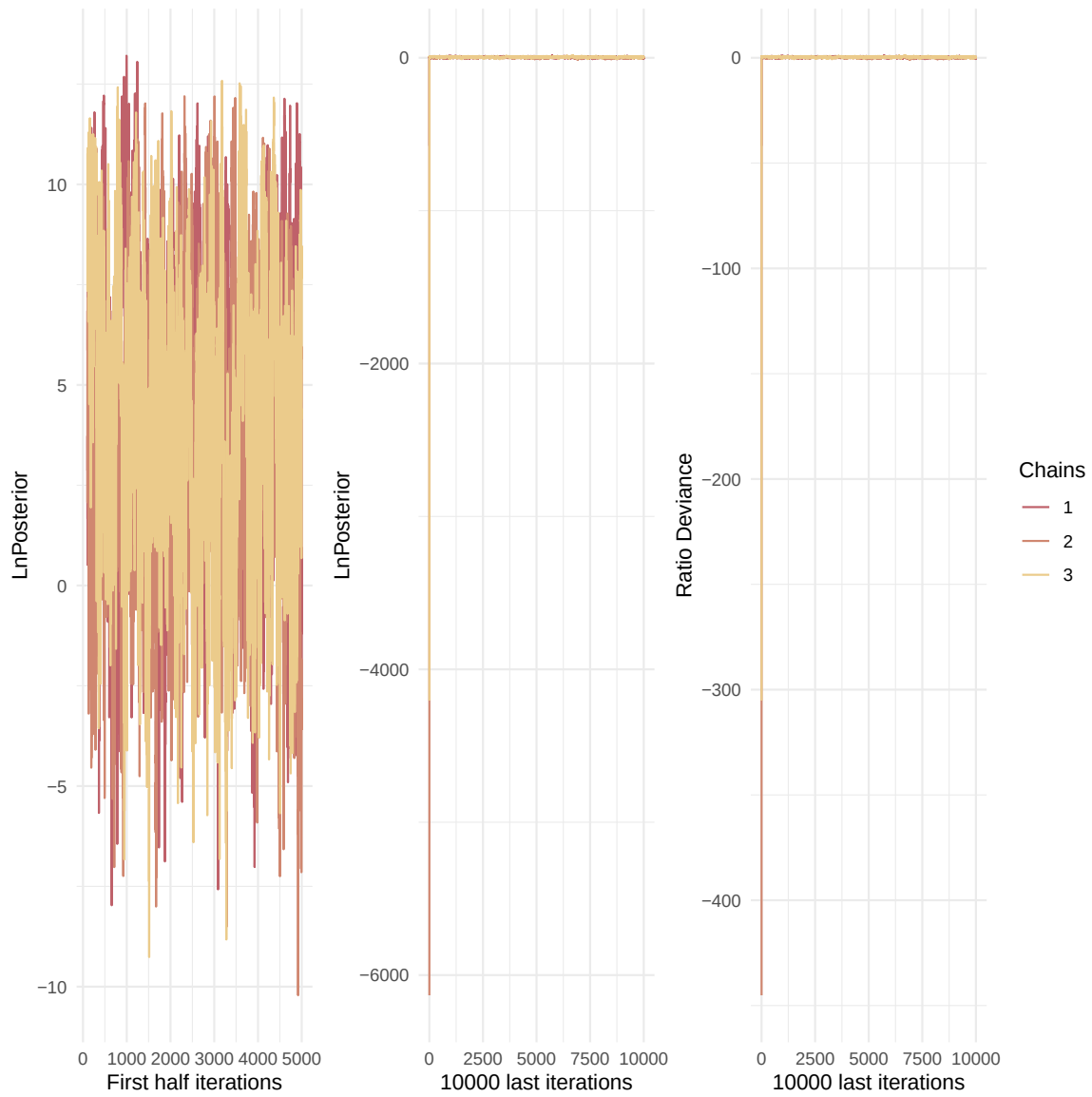


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

kapA.1.	1.25	1.80
pAm.1.	1.74	2.92
r_pAm_pM.1.	1.01	1.03
rOM_ClxHorse.1.	1.02	1.05
Sigma_W.1.	1.00	1.01
v.1.	1.09	1.29
wE.1.	1.51	2.57
z.1.	1.03	1.04

Multivariate psrf

1.53

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

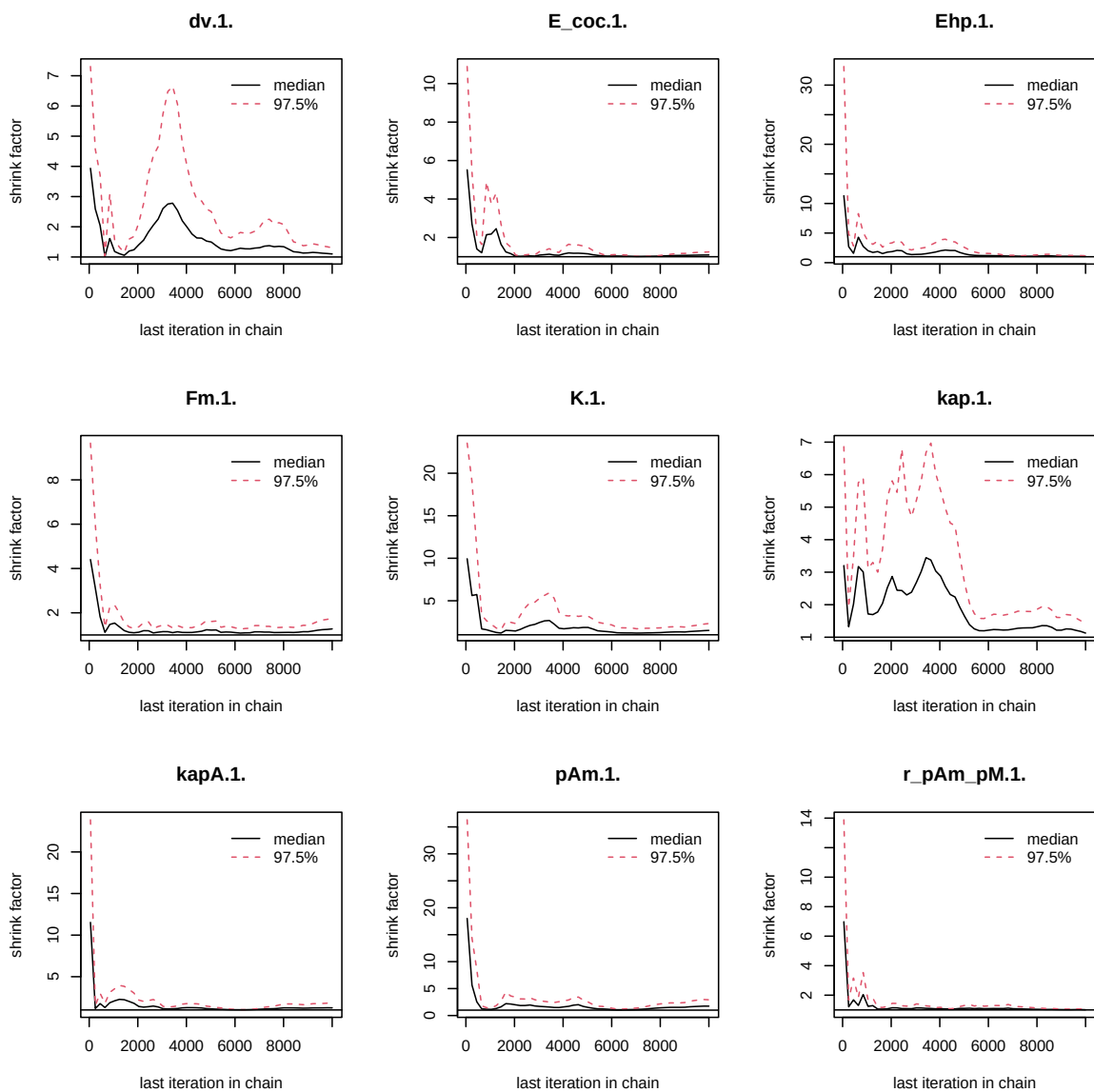


Figure 3: Chains convergence.

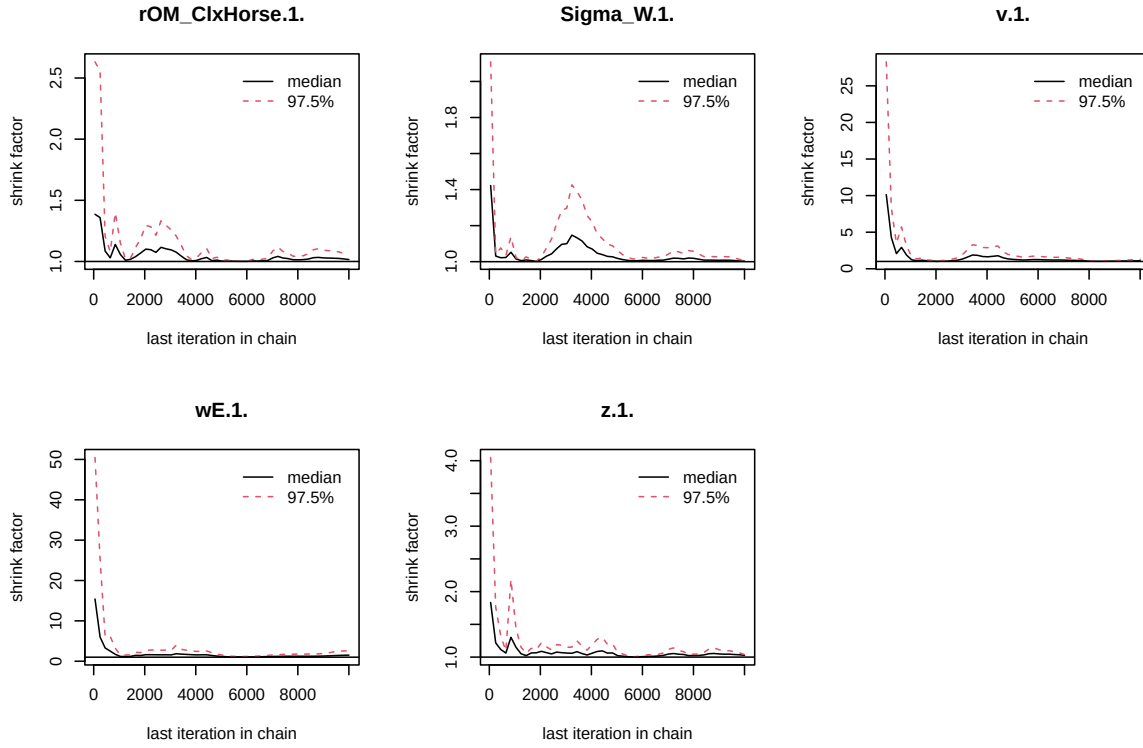


Figure 4: Chains convergence.

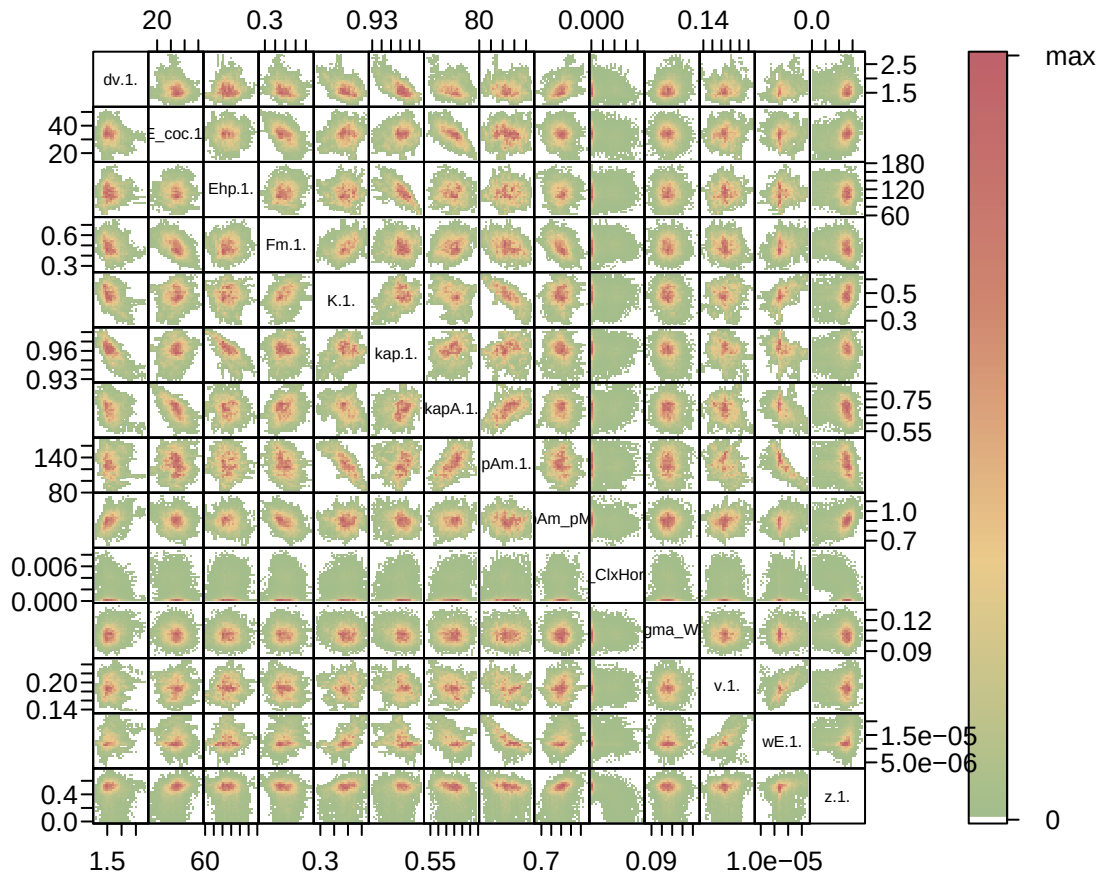


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

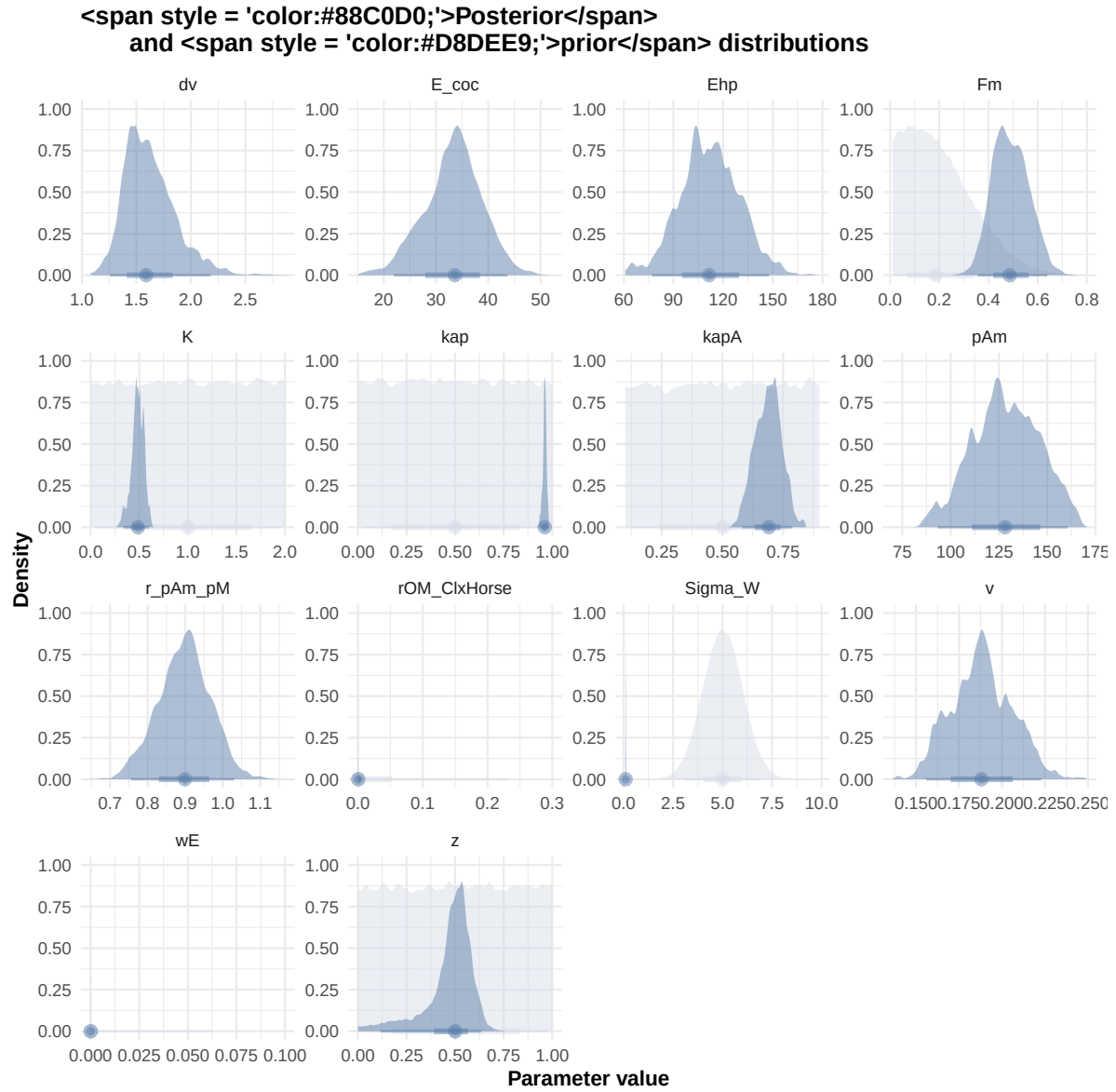


Figure 6: Distributions of the priors and the posteriors.

```
"Bart2020",
#"Gollot2026_lufa"
"Gollot2026_dens_D1",
"Gollot2026_dens_D2",
"Gollot2026_dens_D2b",
"Gollot2026_dens_D5",
"Gollot2026_dens_D7"
)

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\\\.1\\.$|\\.1\\.\"", "", param),
    line = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
  Length_struct,
  Reproduction,
  Energy,
  Organic_matter,"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcali DEB_sim_full.in

#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
```

```
#####  
df_No_sim <- df_data |>  
  dplyr::select(ID_experiment, No_sim)  
  
df_carac_color <- data.frame(  
  carac = unique(l_carac),
```

```

    color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
              Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
  )
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)

```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected relationship between 'x' and 'y'.
 i Row 1 of 'x' matches multiple rows in 'y'.
 i Row 1 of 'y' matches multiple rows in 'x'.
 i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

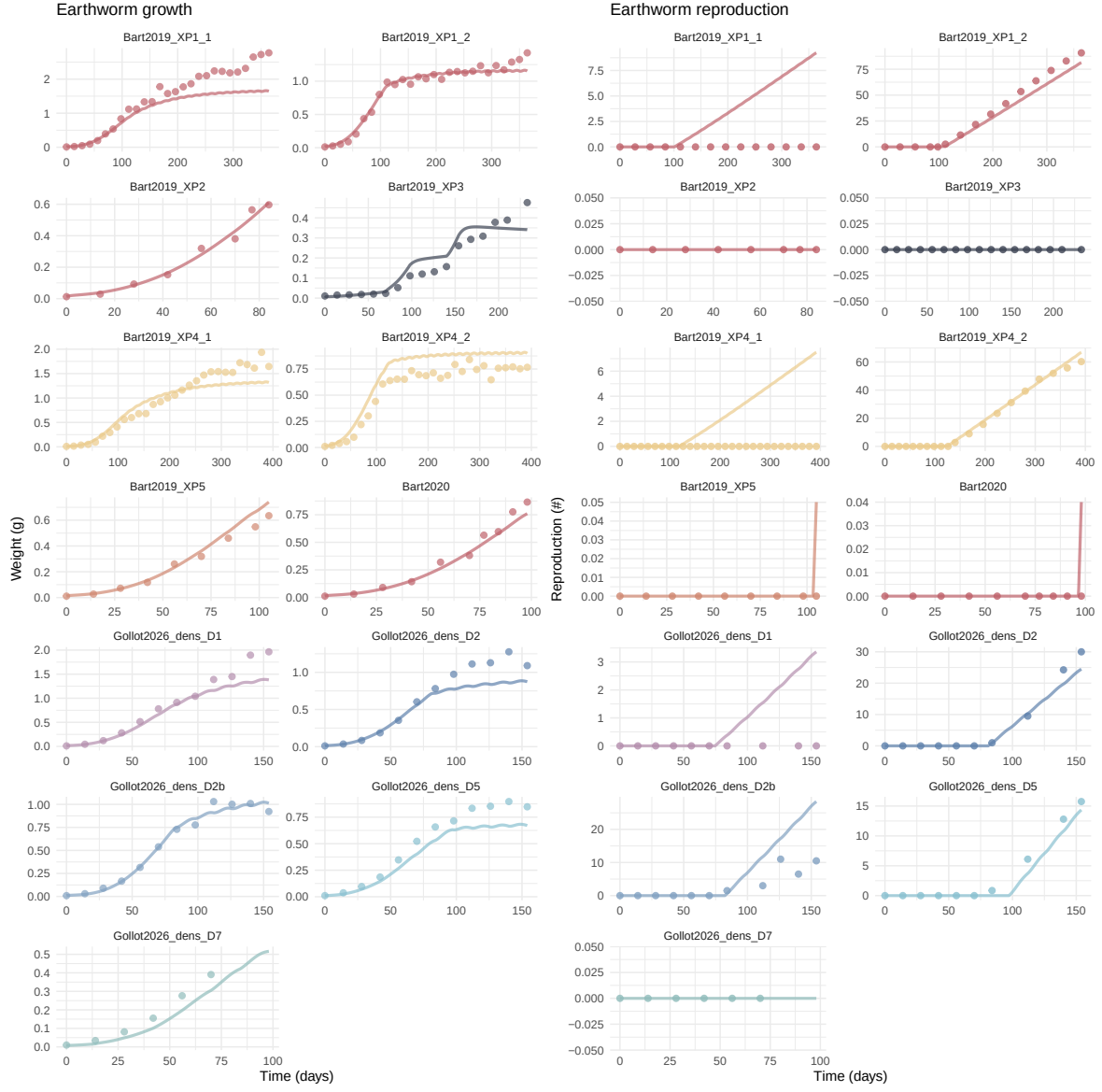
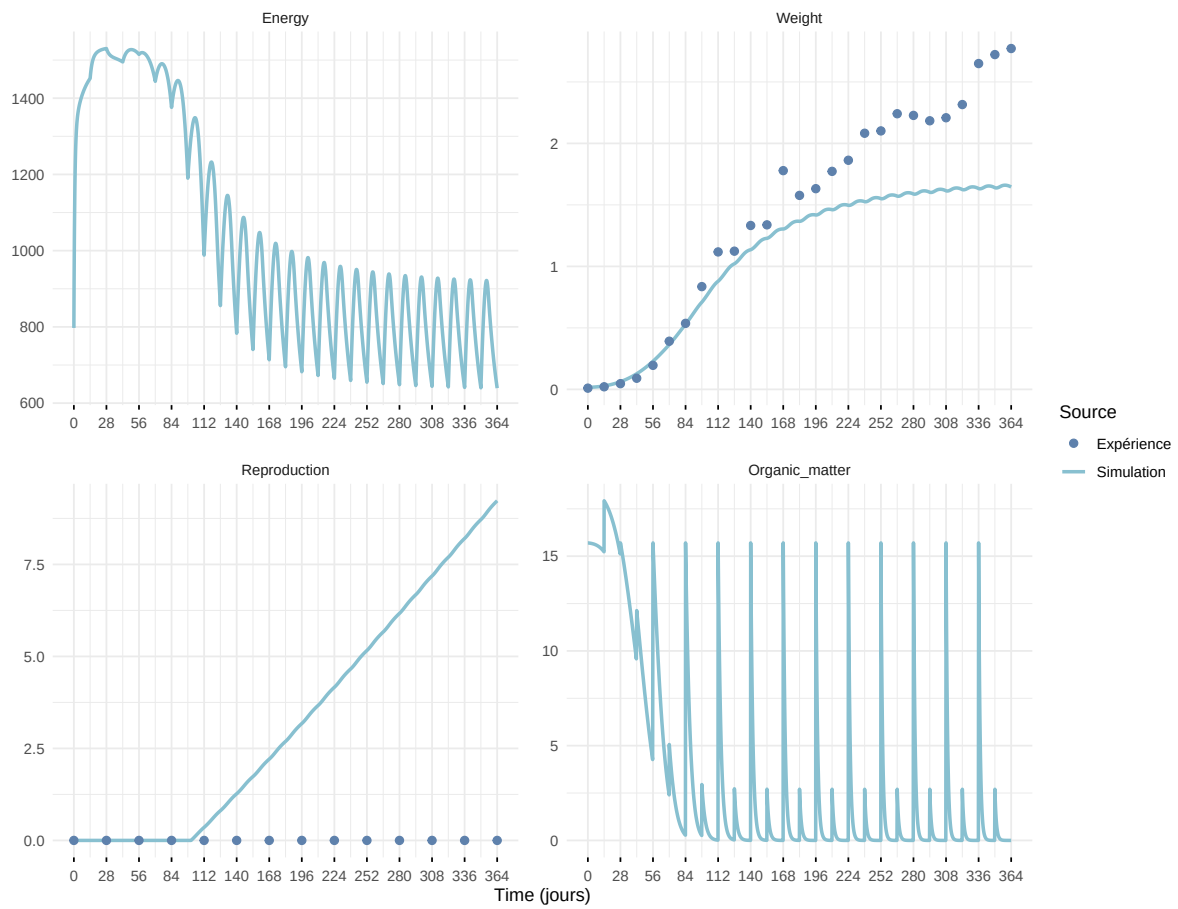
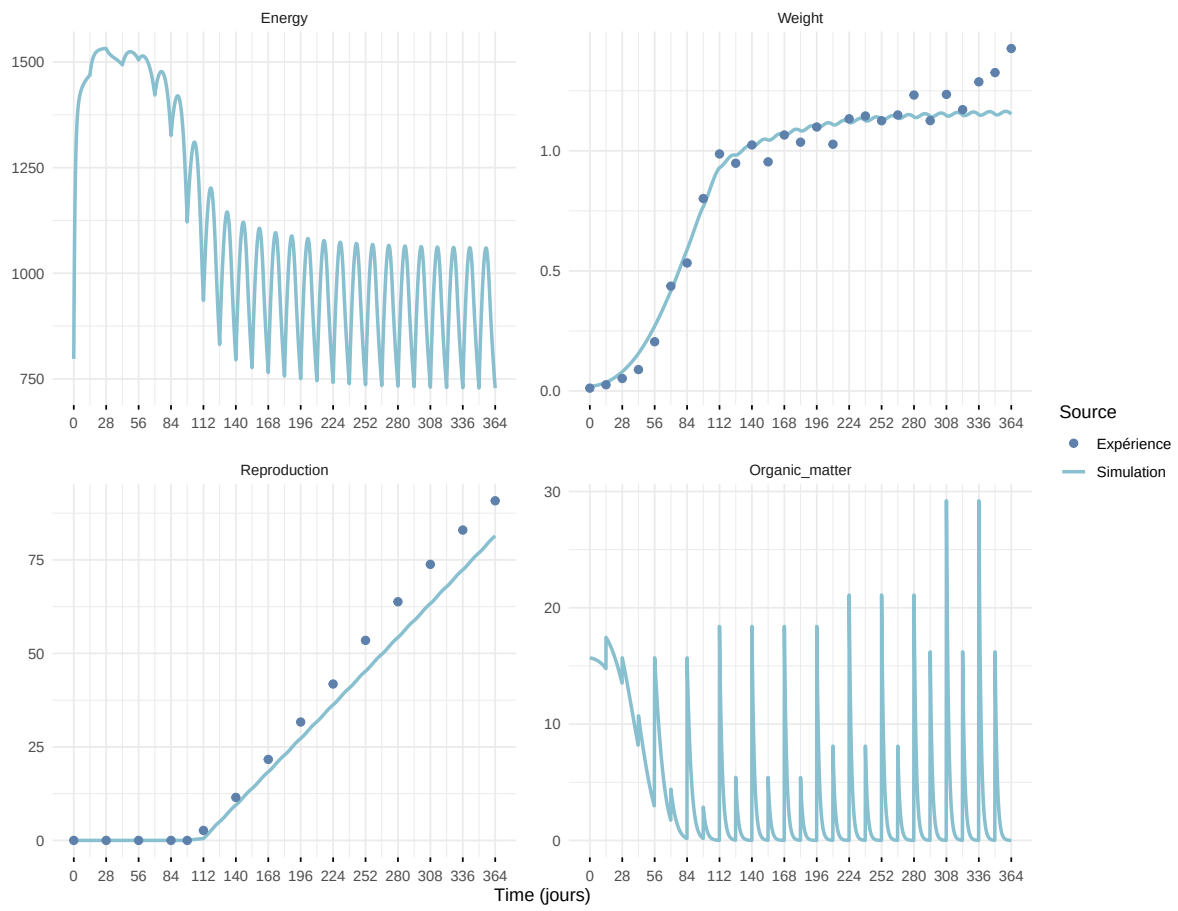


Figure 7: Simulations Weight and Reproduction.

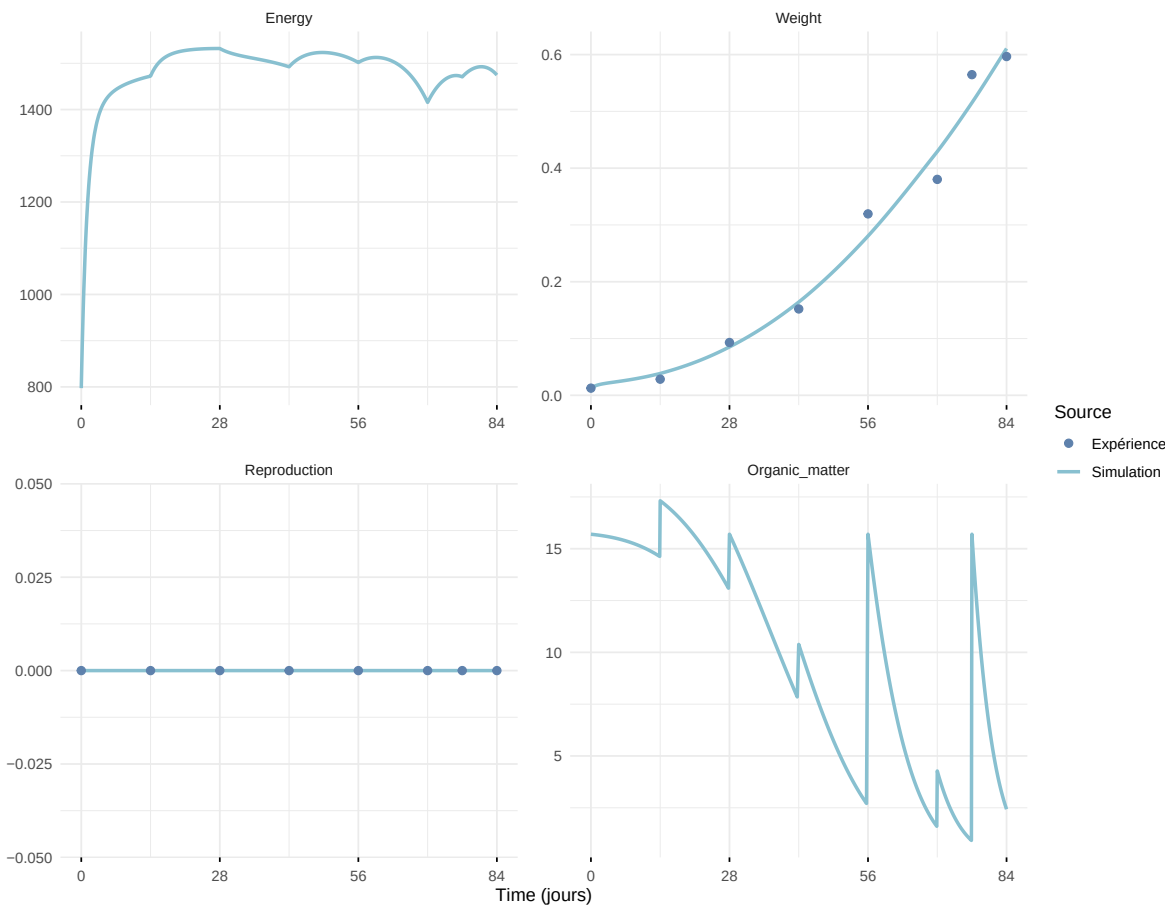
Expérience : Bart2019_XP1_1



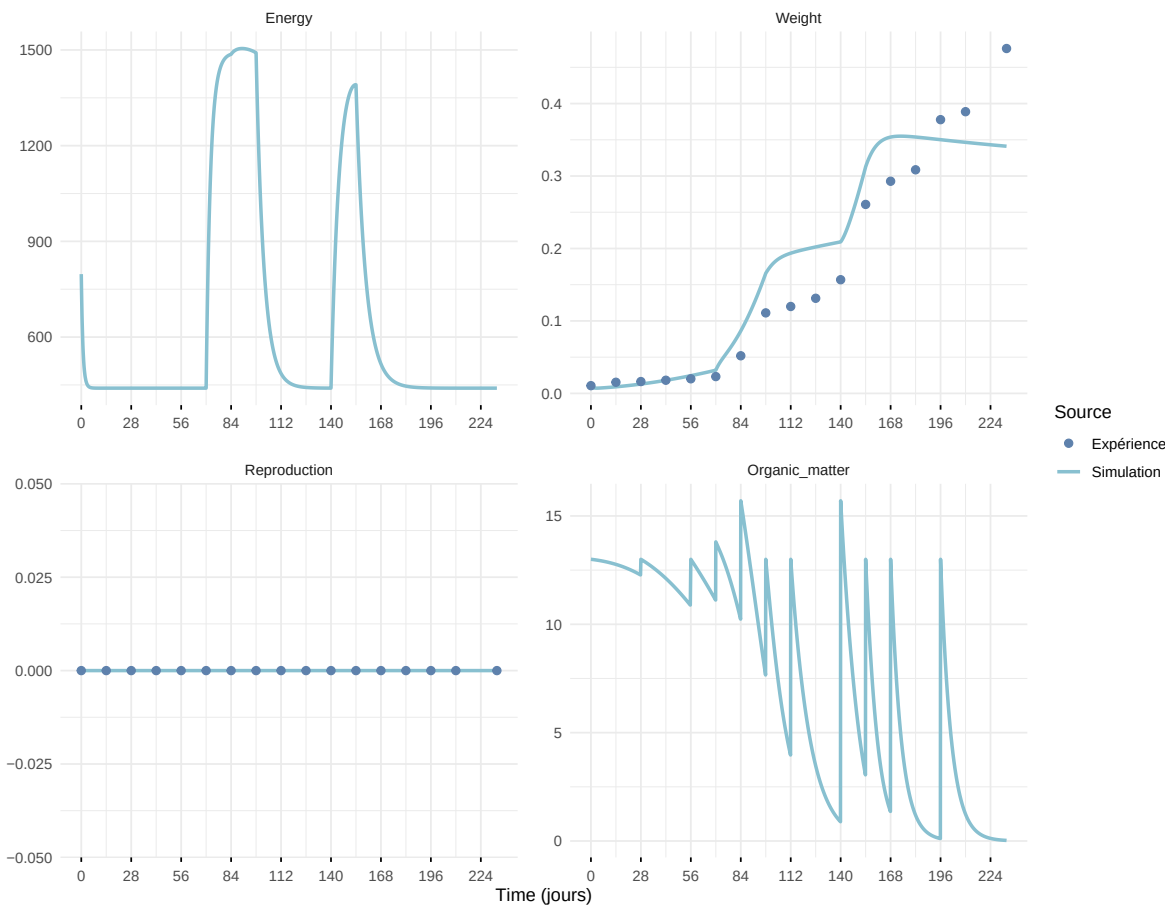
Expérience : Bart2019_XP1_2



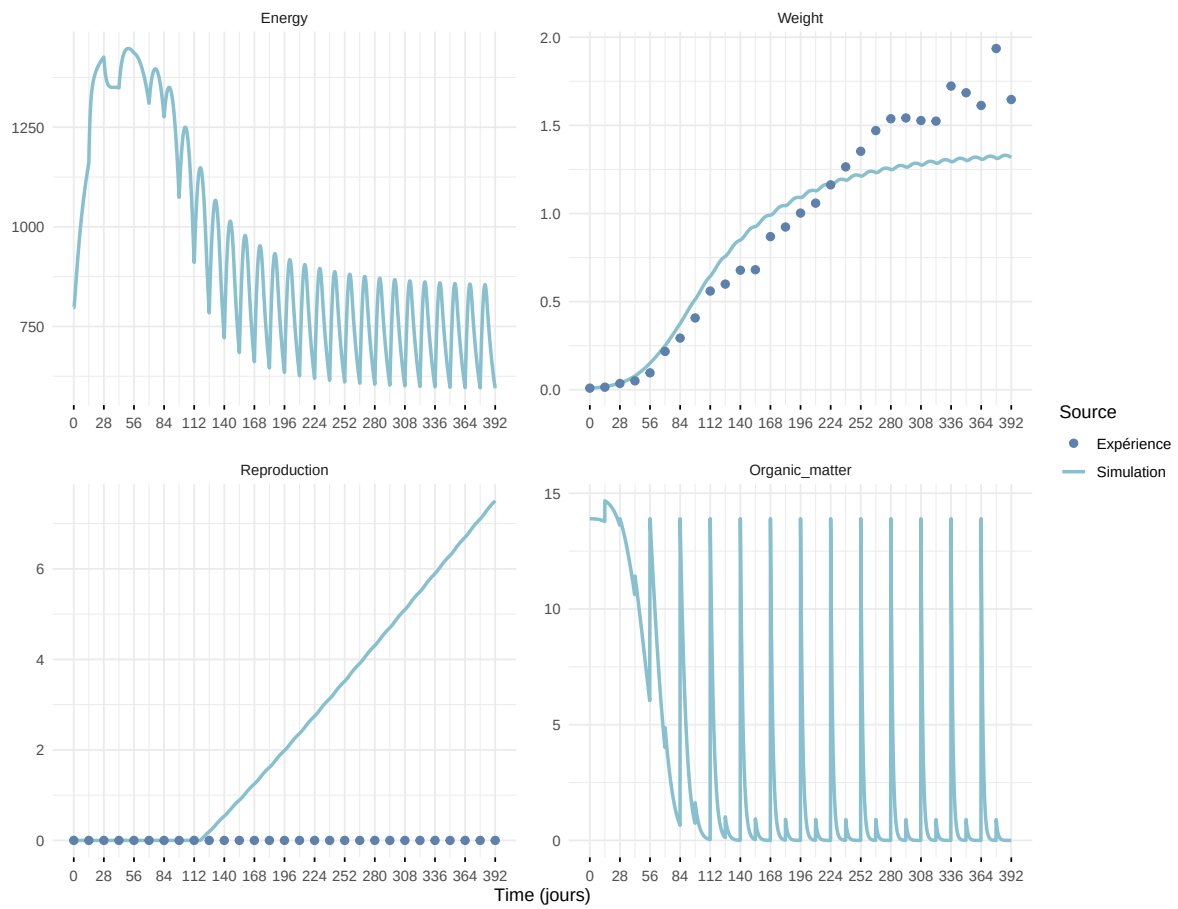
Expérience : Bart2019_XP2



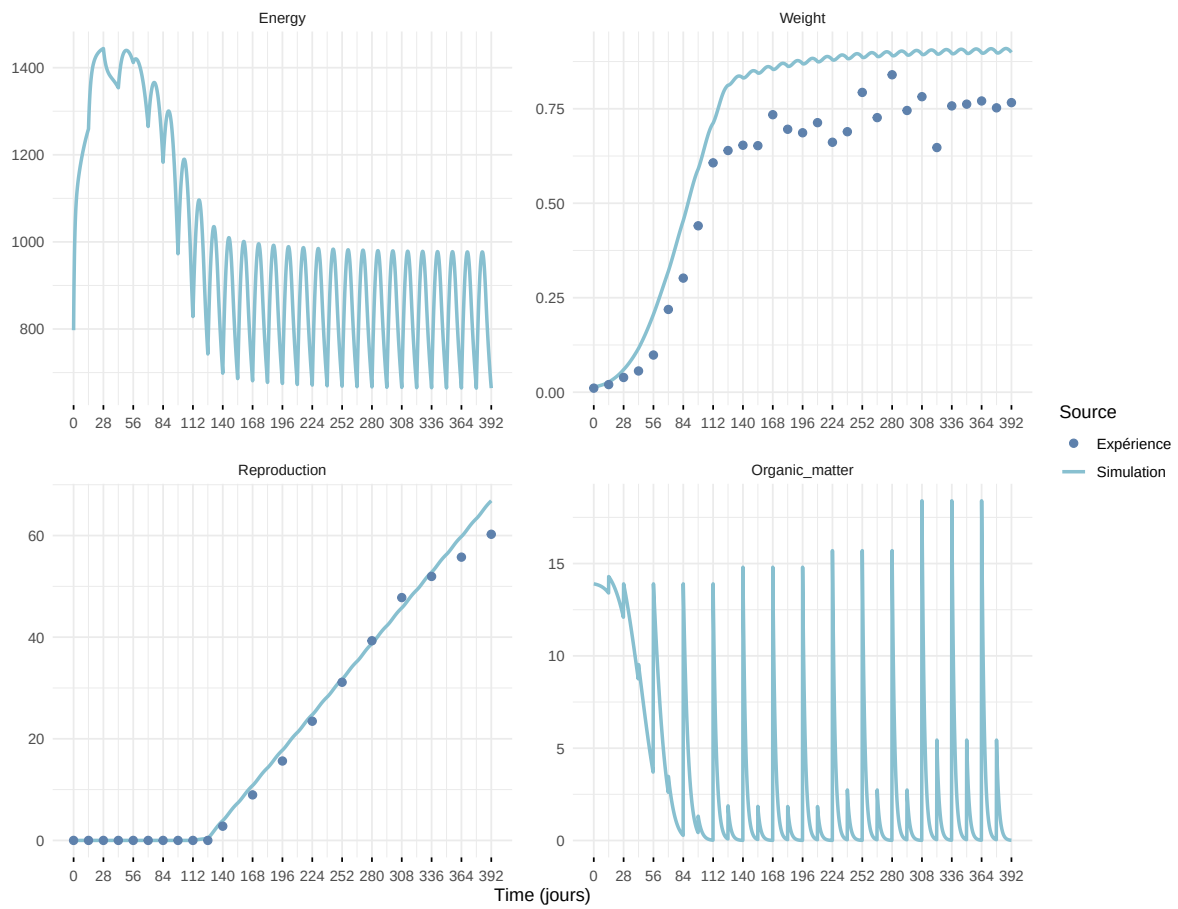
Expérience : Bart2019_XP3



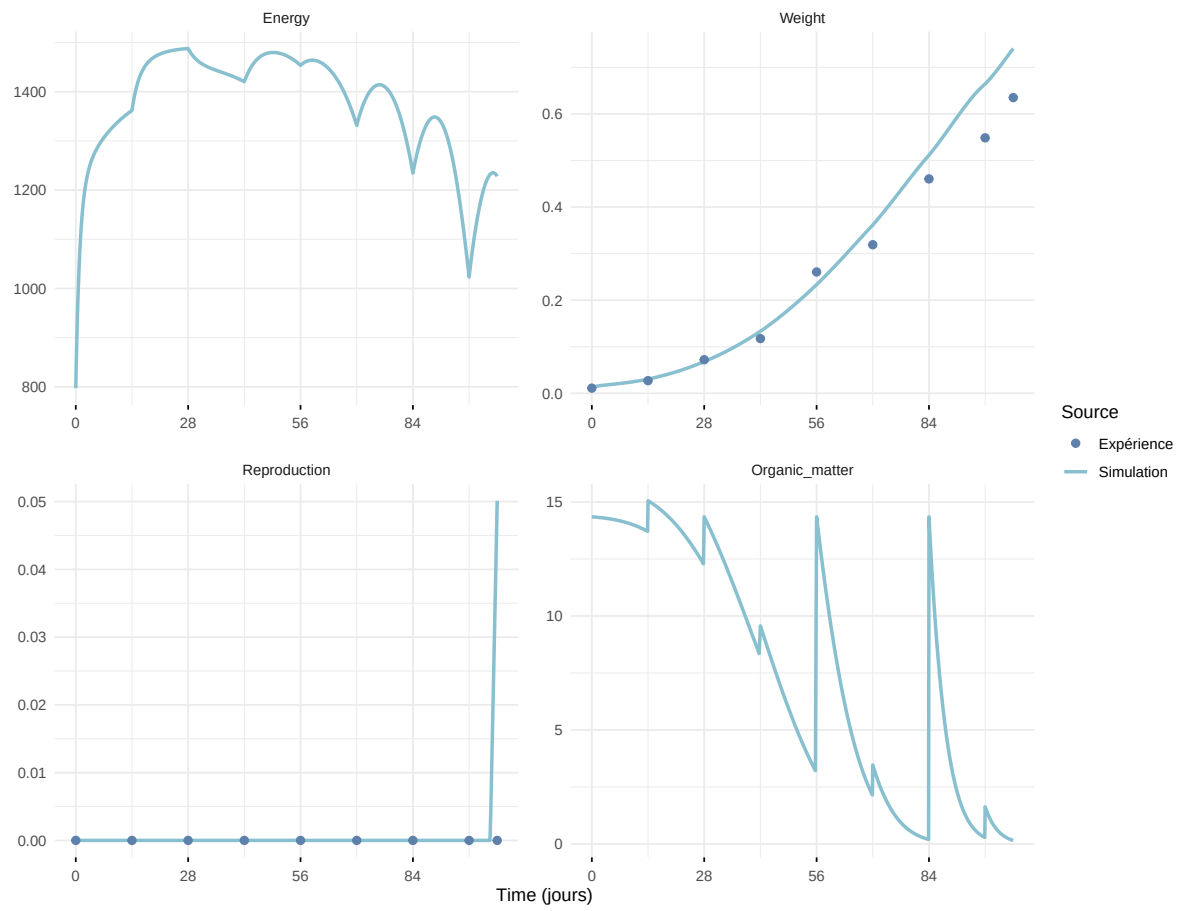
Expérience : Bart2019_XP4_1



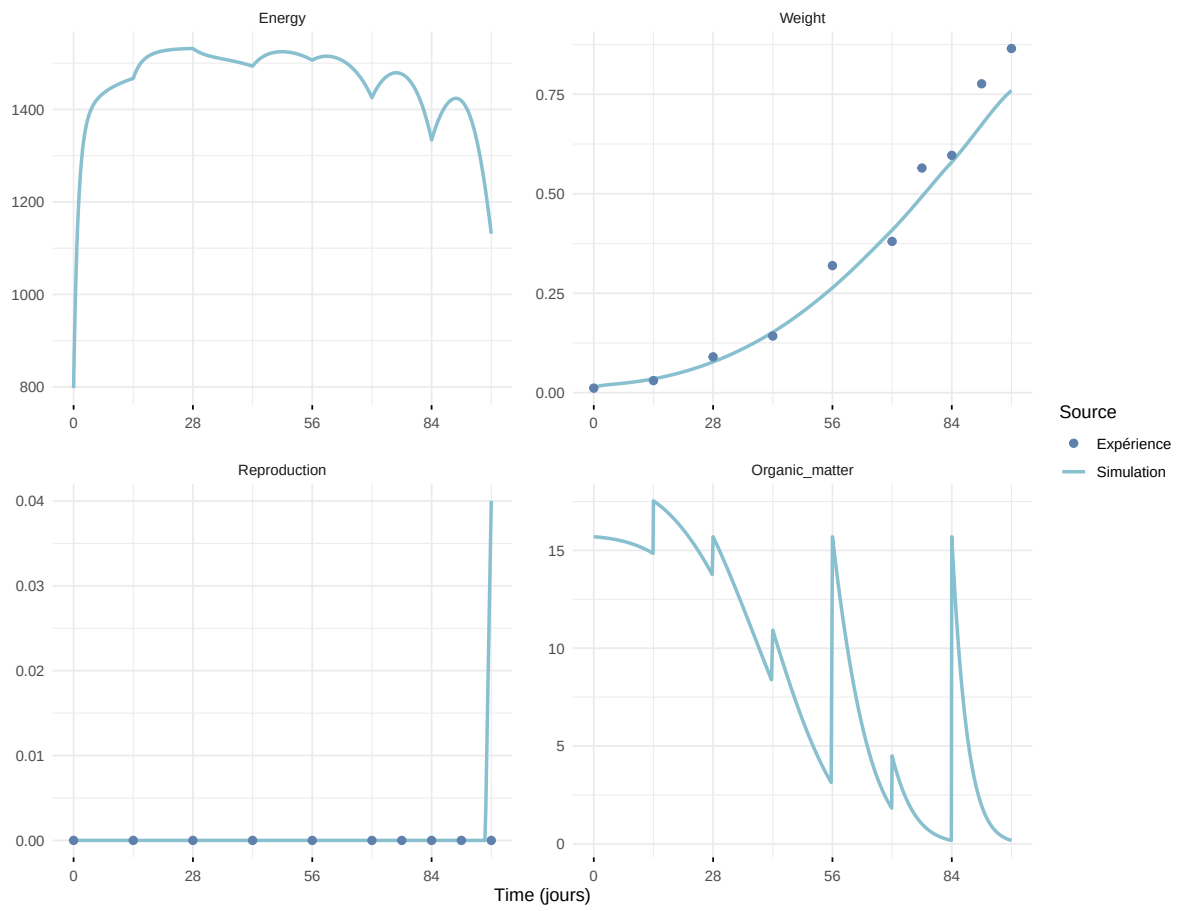
Expérience : Bart2019_XP4_2



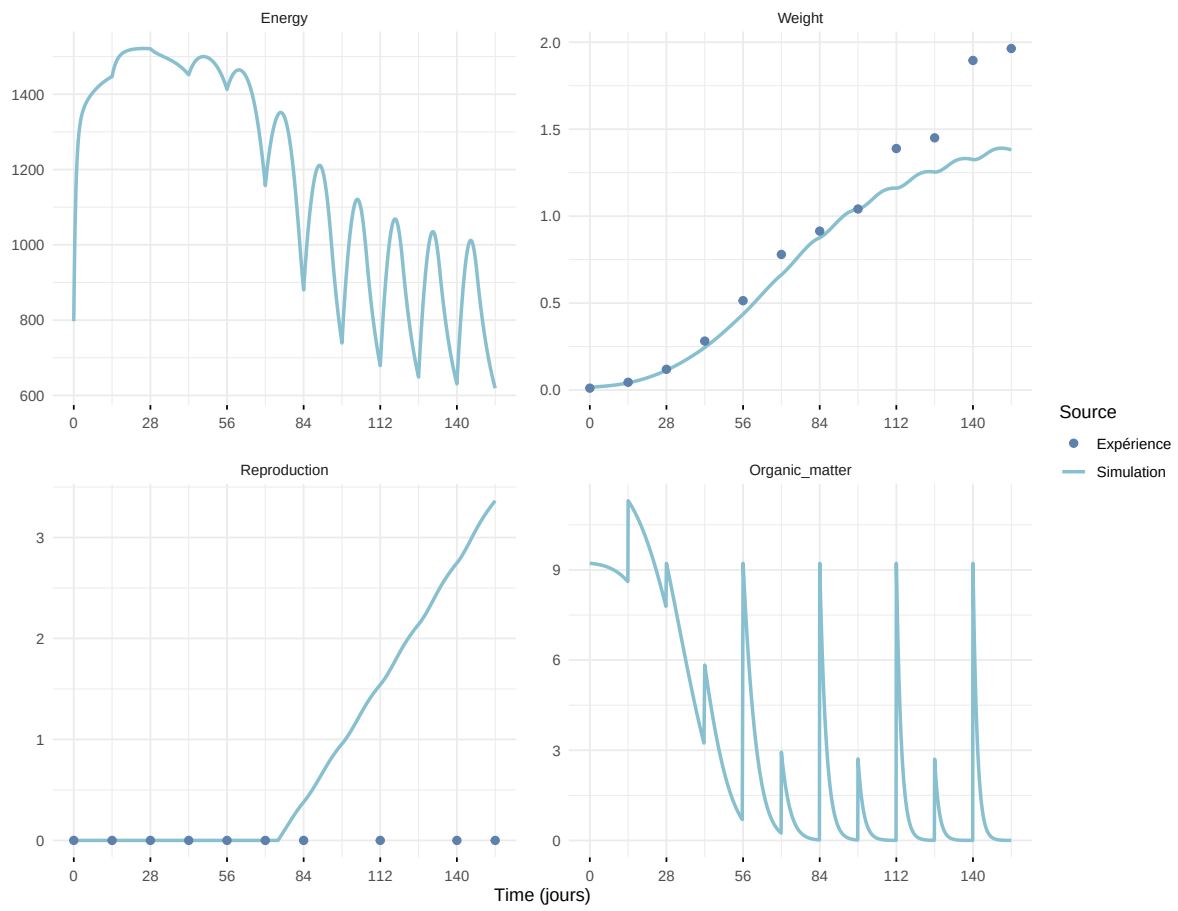
Expérience : Bart2019_XP5



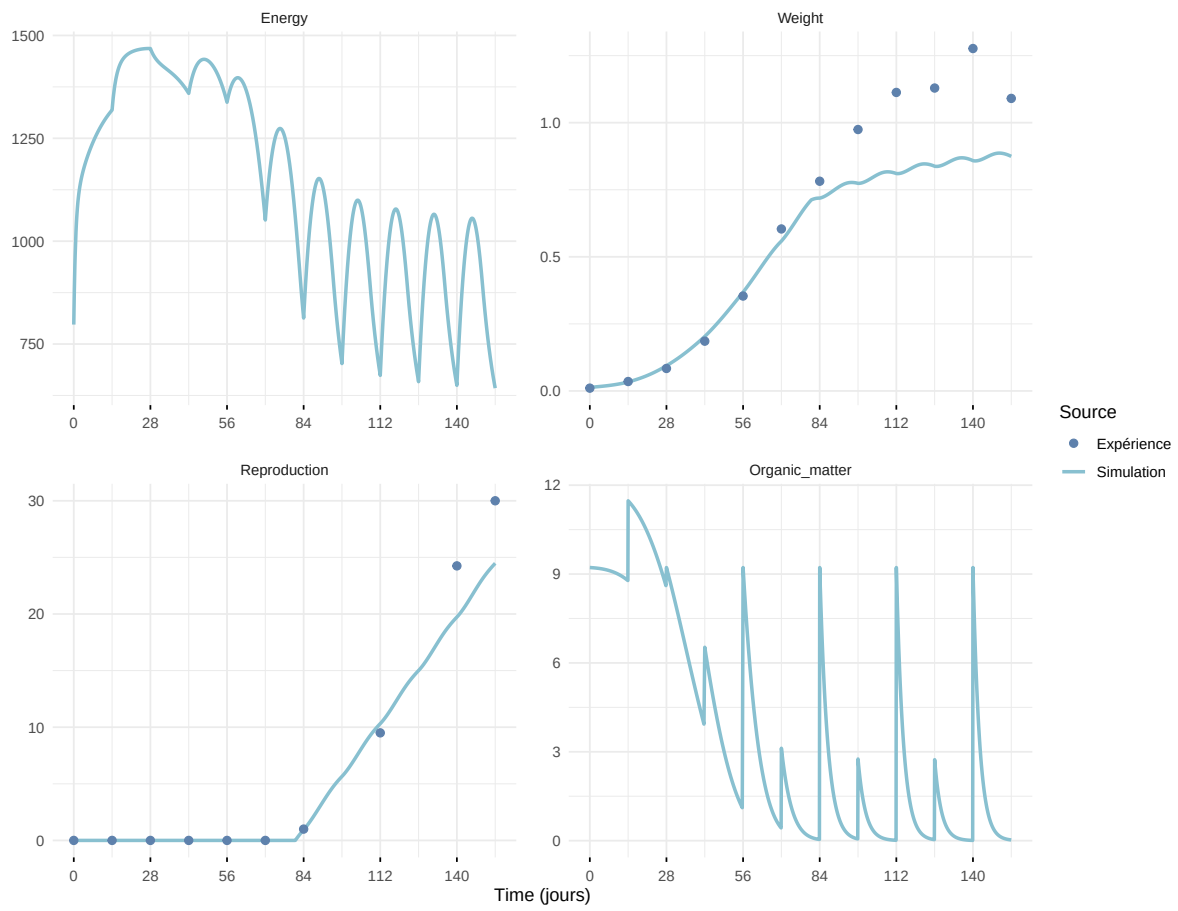
Expérience : Bart2020



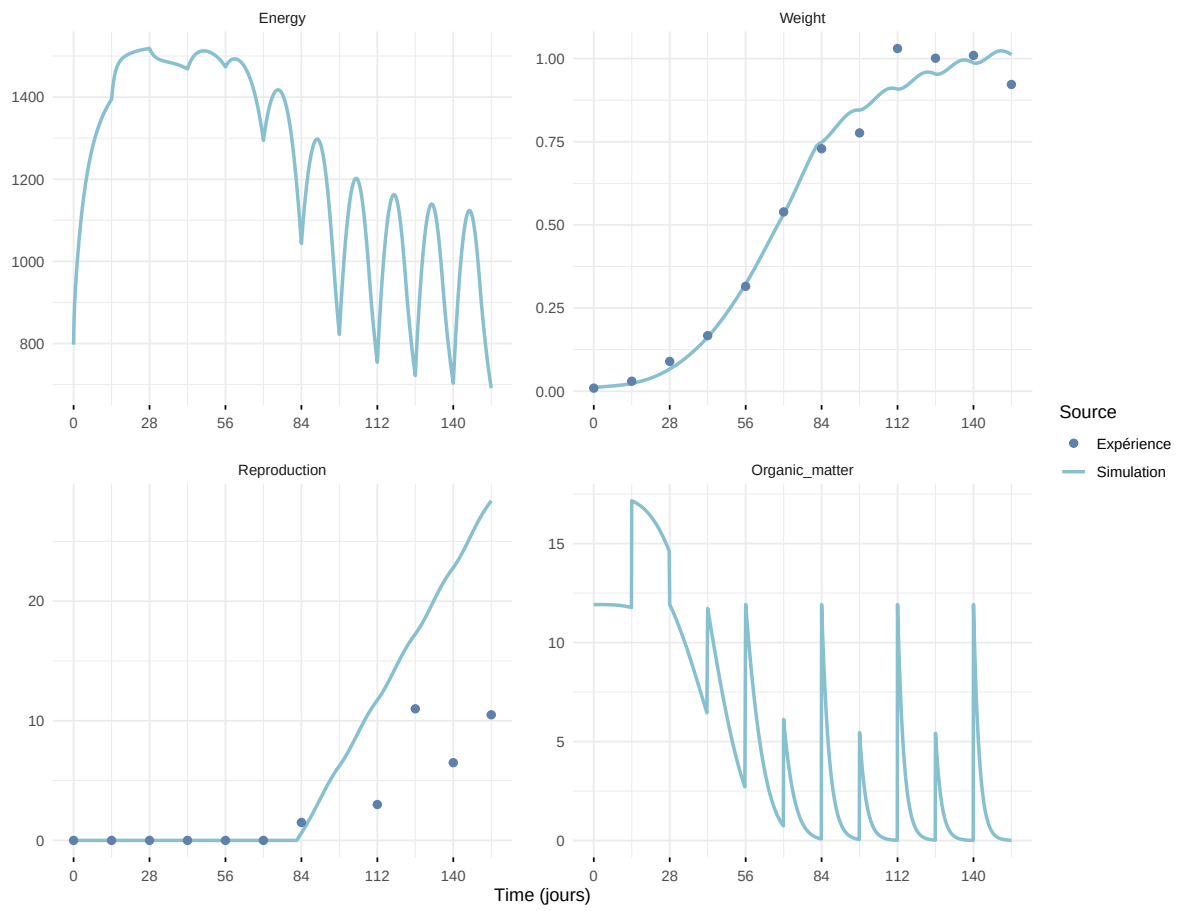
Expérience : Gollot2026_dens_D1



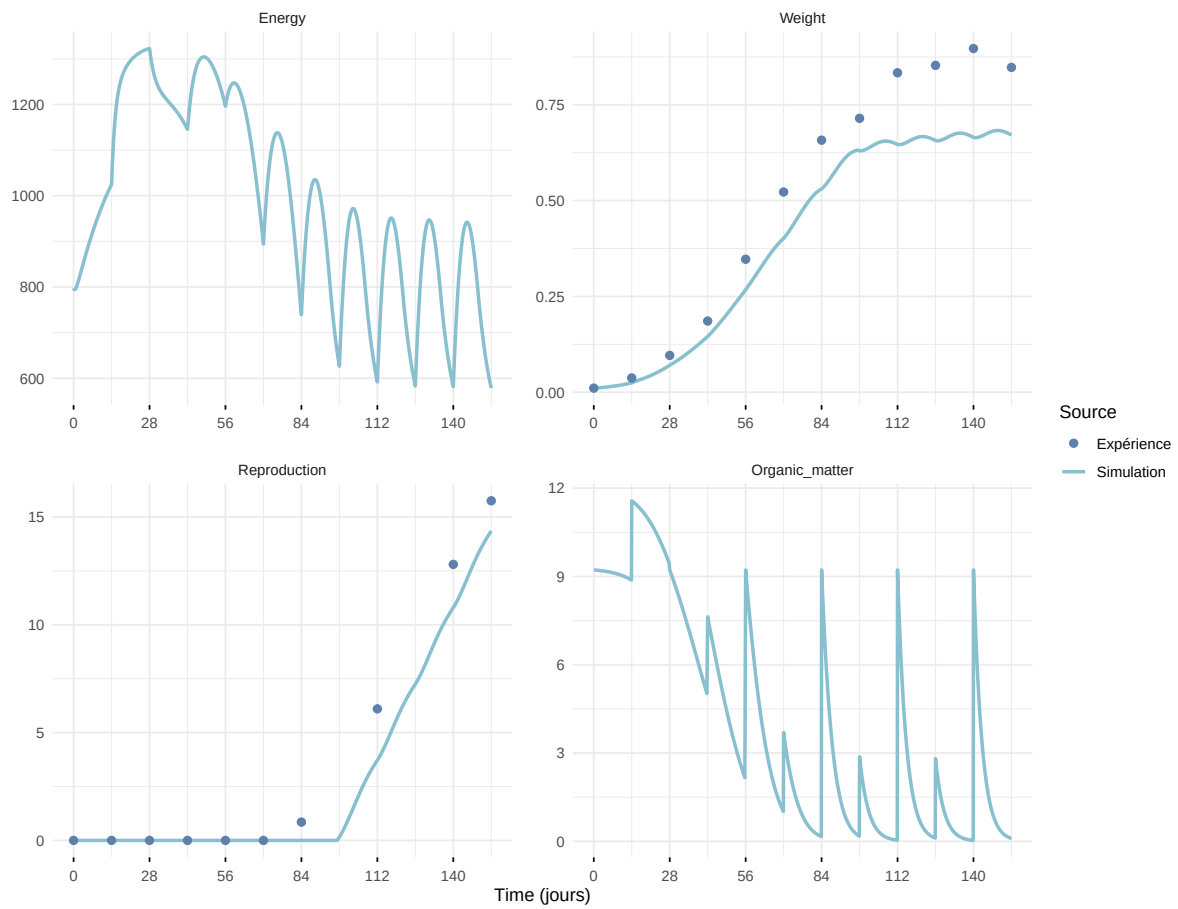
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

